



MANOJ KUMAR BAG | 24MA60R31

COMPUTER SCIENCE AND DATA PROCESSING

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Percentage
2024	M. TECH	IIT Kharagpur	7.73/10
2022	B.ED (MATHEMATICS)	WBUTTEPA	92.15%
2020	M.SC (APPLIED MATHEMATICS)	VISVA BHARATI	66.9%
2017	B. SC (MATHEMATICS)	THE UNIVERSITY OF BURDWAN	55%
2014	HIGHER SECONDARY EXAMINATION	WBCHSE	50%
2012	SECONDARY EXAMINATION	WBBSE	67.71%

PROJECTS

Mitigating Hallucinations in Financial Document QA using Retrieval-Augmented Generation Jan'26 - Apr'26 **Prof. Somesh Kumar**

- Architected a Financial RAG pipeline with a novel HybridRetriever achieving 99% retrieval recall across 100+ benchmark questions.
- Constructed a three-phase retrieval engine combining dense vector search, keyword scanning, and TF-IDF re-ranking for financial QA.
- Achieved 91% overall accuracy and 98% adversarial robustness on annual reports of TCS, Infosys, and Reliance Industries FY2024-25.

Multidocument Summarization Using Transformer Model Jul'25 - Nov'25 **Prof. Somesh Kumar**

- Engineered a preprocessing pipeline for 3M+ news articles using text normalization & entity recognition, reducing input noise by 40%.
- Developed Advanced Hallucination Removal Framework by fine-tuning the Pegasus Transformer model for abstractive summarization.
- Attained state-of-the-art ROUGE-1 score of 47.65 on 11,490 test examples for abstractive summarization, outperforming BART models.

Bitcoin Price Prediction Using Deep Learning Jan'25 - Apr'25 **Prof. Bodhayan Roy, Prof. Buddhananda Banerjee**

- Designed high-performance LSTM and CNN models using TensorFlow, achieving 97.87% prediction accuracy on a 10K+ Bitcoin sample test set.
- Streamlined the data pipeline via automated cleaning, imputation & MinMax scaling, optimizing LSTM with ReLU activation & Adam optimizer.
- Evaluated model reliability using RMSE and MAPE metrics, supported by comprehensive data visualizations to interpret price trend patterns.

Opinion Mining Based Fake Product Review Detection Sep'24 - Oct'24 **Prof. Rupanwita Gayen**

- Processed 40K+ customer reviews using tokenization, stopword removal, stemming to build TF-IDF vectorized features for model training.
- Developed an NLP feature extraction pipeline using TF-IDF, n-grams, and linguistic analysis to identify fraud patterns with 84% precision.
- Trained and optimized Naive Bayes & Random Forest on an Amazon dataset, achieving 86.1% accuracy & 89% recall in detecting fake reviews.

INTERNSHIPS

Market Segmentation using Machine Learning and Data Analysis | Feynn Labs Services Jun'25 - Aug'25 **Mr. Sanjay Basumatary**

- Analyzed 50K+ vehicle sales records across 28 states using K-Means, Hierarchical Clustering, GMM to identify 5 distinct market segments.
- Uncovered 3 high-growth regional markets by feature engineering, Time-Series analysis, and PCA visualization to identify emerging trends.
- Designed a strategic EV entry roadmap & 18 Cr investment plan, projecting 8-12% market share through segment-specific product positioning.

SKILLS AND EXPERTISE

Research & Academic Tools: LaTeX, Jupyter Notebook, Google Cloab, Github | **Programing Languages:** Python, C++, SQL | **ML /Data Science Libraris:** Scikit-Learn, TensorFlow / Keras, NumPy, Pandas, XGboost, Matplotlib, Seaborn | **Core Competencies:** DSA, OOP, DBMS.

COURSEWORK INFORMATION

Computer Science: Object_Oriented Programming | Data Structures & Algorithms | Database Management Systems | Artificial Intelligence & Machine Learning | Programing Language and Data Structures | Computer Organization & Architecture | Parallel Algorithms.
Mathematics: Linear Algebra | Probability & Statistics | Big Data Analysis | Graph Theory & Algorithms | Differential Equations | Convex Optimization.

POSITIONS OF RESPONSIBILITY

Teaching Assistance | Optimization Techniques Lab(MA39207) | Model & Simulation Lab(MA39206)

Under Asst. Prof. Hari Shankar Mahato | Dept. of Mathematics | IIT Kharagpur

Mentoring and guiding 80+ student in the lab, evaluating their lab assignments and conducting examinations.
Served as a freelance Q&A Expert for Chegg India, Providing detailed, explanatory sultions to over 1000 unique math problems.

AWARDS AND ACHIEVEMENTS

GATE 23 & 24 | Qualified the GATE (Mathematics) examination twice, achieved a rank in the top 2.18% (2023).

CERTIFICATIONS

Domestic Data Entry Operator for NSQF level 4 | West Bengal Electronics Industry Devolopment Corporation (Jun'19 - Aug'19).

EXTRA CURRICULAR ACTIVITIES

- Social: Guided 100+ underprivileged students of class 10th, 11th & 12th in studies for 2 year during Covid-19.
- Sports: Part of silver winning Football team in intra Departmental Football Tournament at Visvs Bharati University.